STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Lecture Plan and Review

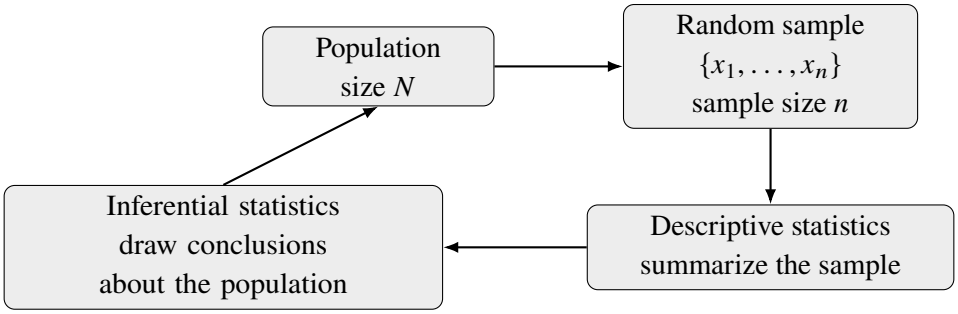
Today:	Sections 6.3 and 6.4.
Next week:	Monday — Section 6.4; Wednesday — Chapter 1.
Last Monday:	Syllabus; course outline; statistics; descriptive statistics; inferential statistics.

1 Section 6.1 — Review: Population and Random Sample

A population has size N. A random sample drawn from that population is written

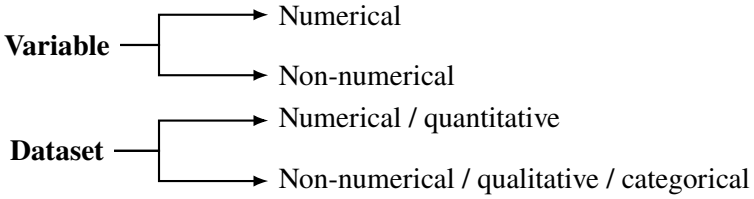
$$\{x_1, x_2, \ldots, x_n\},$$

where n is the sample size.



2 Section 6.2 — Review: Data Types and Graphical Tools

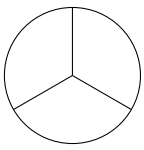
A variable may be numerical or non-numerical; datasets use the corresponding terminology.



For non-numerical datasets, standard displays include bar graphs and pie charts.



(a) Bar graph.



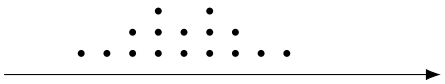
(b) Pie chart.

2.1 Numerical Datasets

For numerical data, common displays include dotplots, stem-and-leaf plots, histograms, and ogives. The basic workflow is

$$\text{Data} \longrightarrow \text{Table} \longrightarrow \text{Graph}.$$

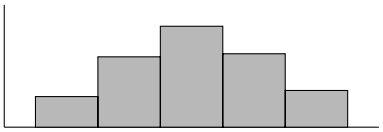
Display	Role
Dotplot	Graphs individual numerical observations.
Stem-and-leaf plot	Organizes small numerical datasets by stem and leaf.
Histogram	Frequency histogram (F.H.) or relative-frequency histogram (R.F.H.).
Ogive	Graph of cumulative relative frequency.



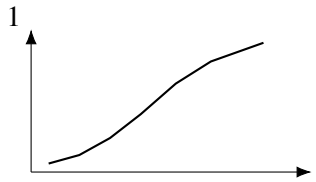
(a) Schematic dotplot.

Stem	Leaf
0	258
1	147
2	03

(b) Schematic stem-and-leaf plot.



(c) Schematic histogram (F.H. or R.F.H.).



(d) Schematic ogive (RCF).

A partial frequency distribution is

Class	Frequency (F)	Relative frequency (RF)	Cumulative frequency (CF)	RCF
$[0, 5)$	1	$1/40$	1	$1/40$
$[5, 10)$	5	$5/40$	6	$6/40$
$[10, 15)$	0	0	6	$6/40$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Final cumulative values			40	1

Homework: 6b; 8(a,d,e); 10(a,d,e); 16.

3 Section 6.3 — Numerical Measures for Numerical Datasets

3.1 I. Measures for the Center

A. Sample Mean/Average

The sample mean is denoted by $\bar{x}$:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}.$$

For the data $\{1, 5, 6, 9\}$,

$$\bar{x} = \frac{1 + 5 + 6 + 9}{4} = \frac{21}{4} = 5.25.$$

The population mean is denoted by μ:

$$\mu = \frac{x_1 + x_2 + \cdots + x_N}{N}.$$

B. Sample Median

The sample median is denoted by $\widetilde{x}$. First, order the dataset.

- If n is odd, $\widetilde{x}$ is the middle value.
- If n is even, $\widetilde{x}$ is the average of the two middle values.

For an odd sample size,

$$\text{ordered data:} \qquad 1, 5, \boxed{6}, 9, 17 \qquad \Longrightarrow \qquad \widetilde{x} = 6.$$

For an even sample size, the data

$$2, 8, 7, 11, 21, 14$$

become

$$2, 7, \boxed{8}, \boxed{11}, 14, 21,$$

so

$$\widetilde{x} = \frac{8 + 11}{2} = 9.5.$$

Resistance to Outliers

The median is a resistant measure: it is not strongly influenced by extreme outliers. The mean is not resistant.

$$\{1, 2, 3, 4, 5\}: \qquad \bar{x} = \widetilde{x} = 3,$$

whereas

$$\{1, 2, 3, 4, 50\}: \qquad \widetilde{x} = 3, \qquad \bar{x} = 12.$$

C. Mode

The **mode** is the value that appears most often in the ordered dataset. Every dataset has one mean and one median, but a dataset may have no mode, one mode, or $2, 3, 4, \ldots$ modes.

3.2 II. Measures for the Variation

A. Range

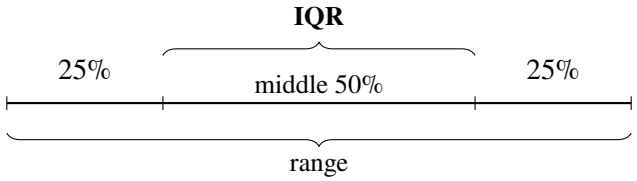
The **range** is the difference between the largest and smallest values in a dataset. For

$$\{8, 11, 9, 13, 5, 21, 34, 27\},$$

$$R = 34 - 5 = 29.$$

B. Interquartile Range — Preview

The interquartile range is, roughly, the range of the middle 50% of an ordered dataset.



A precise definition of IQR follows in Section 6.4.

C. Sample Standard Deviation

The sample standard deviation is denoted by s.

A first attempt might average the signed deviations from the mean, but these deviations cancel:

$$\frac{(x_1 - \bar{x}) + (x_2 - \bar{x}) + \cdots + (x_n - \bar{x})}{n} = \frac{(x_1 + x_2 + \cdots + x_n) - n\bar{x}}{n} = 0.$$

Squaring the deviations gives the definition

$$\boxed{s = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}}.}$$

Expanding the squared deviations gives an equivalent form:

$$s = \sqrt{\frac{(x_1^2 + \cdots + x_n^2) - 2\bar{x}(x_1 + \cdots + x_n) + n\bar{x}^2}{n - 1}}$$

$$= \boxed{\sqrt{\frac{(x_1^2 + x_2^2 + \cdots + x_n^2) - n\bar{x}^2}{n - 1}}},$$

using $x_1 + x_2 + \cdots + x_n = n\bar{x}$.

Example

For

$$\{1, 4, 6, 9, 10\},$$

first compute

$$\bar{x} = \frac{1 + 4 + 6 + 9 + 10}{5} = 6.$$

Using the definition,

$$\begin{aligned} s &= \sqrt{\frac{(1-6)^2 + (4-6)^2 + (6-6)^2 + (9-6)^2 + (10-6)^2}{4}} \\ &= \sqrt{\frac{25 + 4 + 0 + 9 + 16}{4}} = \sqrt{\frac{54}{4}} = \sqrt{13.5}. \end{aligned}$$

Using the alternative formula,

$$\begin{aligned} s &= \sqrt{\frac{1 + 16 + 36 + 81 + 100 - 5(6)^2}{4}} \\ &= \sqrt{\frac{234 - 180}{4}} = \sqrt{13.5}. \end{aligned}$$

In summation notation,

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} \qquad \text{and} \qquad s = \sqrt{\frac{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}{n-1}}.$$

The quantity s^2 is the sample variance. The population variance is denoted by σ^2, and the population standard deviation is

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}.$$

	Size	Mean	Standard deviation
Sample	n	$\bar{x}$	s
Population	N	μ	σ

More variation in a dataset produces a larger standard deviation. For example,

$$\{6, 6, 6, 6, 6\}: \quad \bar{x} = \widetilde{x} = \text{mode} = 6, \qquad R = \text{IQR} = s = 0,$$

$$\{6, 6, 7, 6, 6\}: \quad s_2 > 0, \qquad \{6, 6, 8, 6, 6\}: \quad s_3 > s_2.$$

If each observation x_i is measured in centimeters, then $\bar{x}$, $\widetilde{x}$, and s are measured in centimeters, while s^2 is measured in square centimeters.

Homework: 18 (p. 217).

4 Section 6.4 — Relative Standing, Percentiles, and Quartiles

The topics for Section 6.4 are measures of relative standing, percentiles, quartiles, outliers, and boxplots. The ordered values considered are

$$5,\ 11,\ 12,\ 13,\ 16,\ 21,\ 41.$$

4.1 I. Measures of Relative Standing: z-Scores

A value x in a dataset with mean μ and standard deviation σ has z-score

$$\boxed{z = \frac{x - \mu}{\sigma}}, \qquad \text{equivalently} \qquad \boxed{x = \mu + z\sigma}.$$

The z-score tells how many standard deviations x lies above or below the mean.

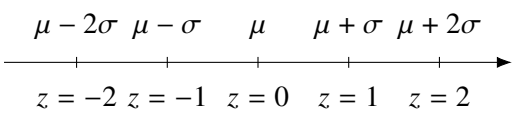
Example

If $\mu = 5$ and $\sigma = 0.5$, then

$$x_1 = 6: \qquad z = \frac{6 - 5}{0.5} = 2,$$

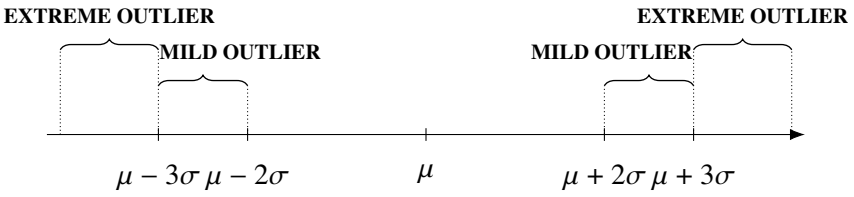
and

$$x_2 = 4.5: \qquad z = \frac{4.5 - 5}{0.5} = -1.$$



The 3-Standard-Deviation Rule

As a rule of thumb, almost all values in a dataset are within three standard deviations of the mean. Equivalently, almost all values have z-scores between -3 and 3.



4.2 II. Percentiles

If the 20th percentile is 8.34, then 20% of the ordered dataset is at or below 8.34, while 80% is at or above 8.34.

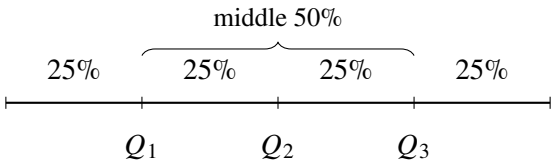
Important percentiles include

$$1\text{st},\ 5\text{th},\ 10\text{th},\ 25\text{th},\ 50\text{th},\ 75\text{th},\ 90\text{th},\ 95\text{th},\ 99\text{th}.$$

4.3 III. Quartiles

The quartiles are the 25th, 50th, and 75th percentiles:

$$Q_1 = \text{first (lower) quartile}, \qquad Q_2 = \text{second quartile}, \qquad Q_3 = \text{third (upper) quartile}.$$



The second quartile is the median of the entire ordered dataset:

$$Q_2 = \widetilde{x}.$$

The first quartile Q_1 is the median of the lower half, and the third quartile Q_3 is the median of the upper half. The interquartile range is

$$\boxed{\text{IQR} = Q_3 - Q_1}.$$

Finding the Quartiles: Even n

If n is even, the lower half consists of the smallest 50% of the values and the upper half consists of the largest 50%.

For the ordered data

$$1, 5, 6, 9, 10, 18,$$

the halves are

$$\underbrace{1, 5, 6}_{\text{lower half}} \qquad \underbrace{9, 10, 18}_{\text{upper half}}.$$

Thus

$$Q_2 = \widetilde{x} = \frac{6+9}{2} = 7.5, \qquad Q_1 = 5, \qquad Q_3 = 10,$$

and

$$\boxed{\text{IQR} = Q_3 - Q_1 = 10 - 5 = 5}.$$

Finding the Quartiles: Odd n

For the odd-n example, the median is included in both halves. With

$$2, 8, 9, 10, 17, 18, 35,$$

we have

$$Q_2 = \widetilde{x} = 10,$$

and the two halves are

$$\underbrace{2, 8, 9, 10}_{\text{lower half}} \qquad \underbrace{10, 17, 18, 35}_{\text{upper half}}.$$

Therefore, $Q_1 = 8.5$ and $Q_3 = 17.5$.